TRACE

Typed Reasoning with Attribution, Certificates, and Evidence

A Unified Framework for Evidence-Grounded Reasoning across Formal Substrates

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STRUCTURED INFERENCE / VERIFIABLE OUTPUTS / CONTROLLED TRANSFER

Reasoning succeeds when a structured obligation is discharged by valid evidence. TRACE couples typed construction, budgeted search, and missing-support diagnosis through a shared obligation graph, exposing both the reason for a decision and the conditions under which an improvement transfers.

CONSTRUCT → CHECK → DIAGNOSE → REPLAY
one state representation • native verifier semantics

01 The TRACE engine: a typed, evidence-carrying obligation network

BUDGETED POLICY
$a_t = \pi(G_t, b_t, v_t)$
$b_{t+1} = b_t - \text{cost}(a_t)$

EXPAND | RETRIEVE | ASSEMBLE | REPAIR | STOP

Whole-task clock • attempt-level traces • matched initial state • verifier cost included

FIXED TOTAL COMPUTATION
base attempt → structural expansion → targeted repair

$G = (V, E, \tau, s, \text{provenance})$
$\Sigma\ \text{cost}(a_t) \le B$

CONTEXT → OBLIGATIONS

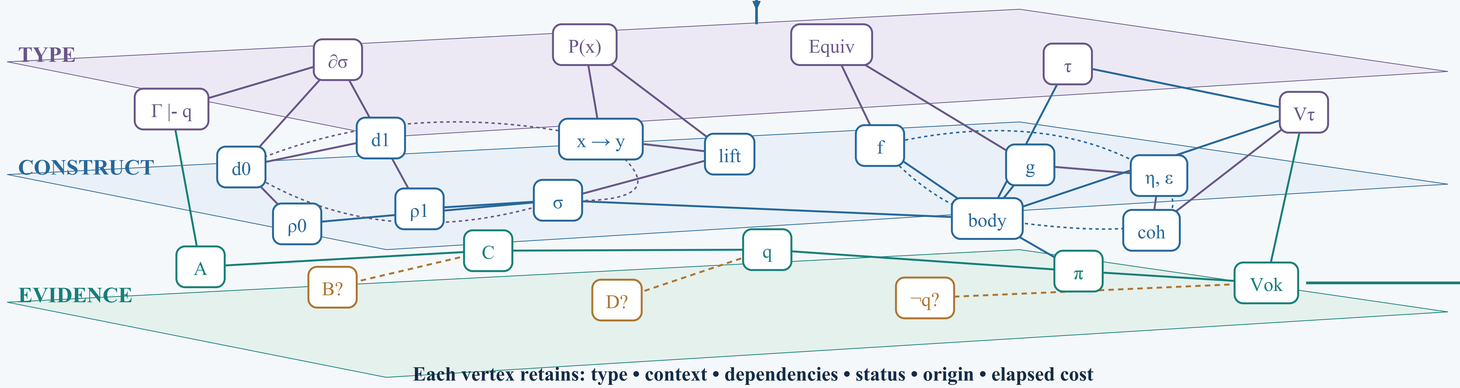
Lean theorem
Γ |- q; local hypotheses
tactic state → subgoals

Open-world theory
T = facts + rules; query q
positive + opposite closure

Rzk construction
Γ |- body : signature
boundary / lift / witness

ONE STATE • THREE COUPLED RELATIONS

dependency E^d / typing E^t / evidence E^e



Each vertex retains: type • context • dependencies • status • origin • elapsed cost

NATIVE VERIFICATION
Lean kernel • closure • rzk typecheck

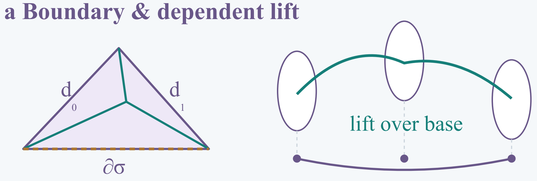
CERTIFIED
proof of q; derivation of ¬q; typed body

ABSTAIN + WITNESS
neither polarity follows; support is absent

UNRESOLVED
budget exhausted or checker rejection

Search failure ≠ logical Unknown.

a Boundary & dependent lift



Compatibility: $d_i(d_j\sigma) = d_{j-1}(d_i\sigma),\ i < j$

b Missing-support certificate

T: A; A & B → C; C → q; D → q
cl(T) contains A; B and D are absent.
q not in cl(T) AND ¬q not in cl(T)

w = {blocked r1: B; blocked r3: D}

Witness is checked against the theory, not confidence.

c Typed witness assembly

Equiv(A, B)
f | is-equiv g, η, ε

body := (f, (g, (η, ε)))
Γ |- body : Equiv(A, B)
splice → typecheck → attach result
ill-typed subterm → obligation ID

A correct map alone is not a complete equivalence.

d Counterfactual trace replay

Freeze z = (context, candidate, feedback, B)

z + baseline instruction → y0

z + structural instruction → y1

Δ = mean[valid(y1) − valid(y0)]

Same state, same checker; only instructions change.

02 Experimental evidence: computation, certificates, and structural reach

A Compute attribution and host dependence

Pass@1 / 60	Single	Portfolio	Δ pp
Host validation	28.3%	36.7%	**+8.3**
Host test	31.7%	35.0%	**+3.3**
ReProver validation	45.0%	38.3%	**−6.7**
ReProver test	40.0%	38.3%	**−1.7**

TIME UTILIZATION AT NOMINAL B = 60 s

Single — **19.2% ≥55 s**
Portfolio — **56.7% ≥55 s**

Matched-compute single: 15% → 30%. Retrieval removal preserves the local gain; changing the prover reverses it.

B Evidence quality under missing support

Joint (%) / 3 seeds	In-domain	Depth 5
Proof-only	39.9 ± 0.6	**26.5 ± 1.1**
ProCo	75.0 ± 0.7	**55.0 ± 0.7**
Gain	+35.1 pp	**+28.5 pp**

SUPPORT-DELETION AUDIT / 4,000 MUTANTS

2,017.7 ± 40.6 accepted missing-support witnesses

Strict Unknown acceptance: 71.9% ID / 63.3% depth 5.

4,096 training questions per seed; fixed 4,000-example evaluation. Gains concern evidence quality, not raw classification.

C Structural recovery and shared failures

Structural comparison	Reference	Intervention
28 held-out: direct → Lite	9 / 28	**14 / 28**
120 fixed failures: replay	5 / 120	**11 / 120**
437 broad: direct → repaired	129 / 437	**154 / 437**

REPAIRED 8-MODEL UNION / 437 TASKS



197 covered / 45.1% **240 shared misses**

88 failures: witness 31; boundary 20; target 16; lift 14; kind 7.
Model union is retrospective coverage.

SEMANTIC CONTRACT
Native checks separate unsupported queries from incomplete search.

ATTRIBUTION CONTRACT
Replay fixed states; isolate guidance, retrieval, repair, and total cost.

TRANSFER CONTRACT
Test prover, depth, and family; no joint TRACE benchmark is reported.

github.com/ChouYuanjue/lean-portfolio-attribution
github.com/ChouYuanjue/proof-countermodel-distill
github.com/ChouYuanjue/rzk-ai4math